# SC12 - Revision H purchasing BOM

140 fitted parts: 129 factory SMT + one user-soldered MCU + 10 user-fitted through-hole parts. Three loose caps extra. Parts estimate: $27.08/board at 10 boards before SMT attrition and spare MCUs; PCB, assembly, freight and tax extra. Unbuilt prototype; check current sourcing and release status before ordering.

| Qty | MPN / supplier part | References | Value / package | Extended USD first tier |
|---|---|---|---|---|
| 1 | 0451003.MRL / C13982 | F1 (Factory SMT) | 3A FAST / Fuse_Littelfuse-NANO2-451_453 | 0.2556 |
| 4 | 0805W8F1001T5E / C17513 | R4, R110, R210, R310 (Factory SMT) | 1k / R_0805_2012Metric | 0.0168 |
| 12 | 0805W8F1002T5E / C17414 | R3, R21, R22, R40, R41, R105, R205, R305, R52, R71, R75, R31 (Factory SMT) | 10k / R_0805_2012Metric | 0.0468 |
| 8 | 0805W8F1003T5E / C149504 | R50, R111, R112, R211, R212, R311, R312, R70 (Factory SMT) | 100k / R_0805_2012Metric | 0.0472 |
| 1 | 0805W8F1152T5E / C17418 | R61 (Factory SMT) | 11.5k / R_0805_2012Metric | 0.0041 |
| 3 | 0805W8F2203T5E / C17556 | R101, R201, R301 (Factory SMT) | 220k / R_0805_2012Metric | 0.015 |
| 1 | 0805W8F3300T5E / C17630 | R30 (Factory SMT) | 330 / R_0805_2012Metric | 0.0054 |
| 1 | 0805W8F5231T5E / C17739 | R51 (Factory SMT) | 5.23k / R_0805_2012Metric | 0.0023 |
| 1 | 0805W8F5902T5E / C17763 | R10 (Factory SMT) | 59k / R_0805_2012Metric | 0.0023 |
| 1 | 1206W4F1200T5E / C17909 | R42 (Factory SMT) | 120 / R_1206_3216Metric | 0.0074 |
| 3 | 1824420000 / 281-2020-ND | J1, J2, J3 (Hand solder) | 2P PUSH 3.5mm / Weidmuller_LSF_90_2P | 4.89 |
| 3 | 1824430000 / 281-2021-ND | J4, J5, J6 (Hand solder) | 3P PUSH 3.5mm / Weidmuller_LSF_90_3P | 6.63 |
| 1 | ABM3B-8.000MHZ-B2-T / C83312 | Y1 (Factory SMT) | 8MHz 18pF / Abracon_ABM3B_5.0x3.2mm | 0.5892 |
| 3 | AO3401A / C15127 | Q101, Q201, Q301 (Factory SMT) | AO3401A / SOT-23 | 0.2829 |
| 1 | AO4485 / C51499 | Q1 (Factory SMT) | AO4485 / SOIC-8_3.9x4.9mm_P1.27mm | 0.2951 |
| 1 | AP2112K-3.3TRG1 / C51118 | U2 (Factory SMT) | AP2112K-3.3 / SOT-23-5 | 0.1712 |
| 2 | B240A-13-F / C92506 | D4, D5 (Factory SMT) | B240A / D_SMA | 0.1902 |
| 3 | BAT46W-E3-18 / C5557780 | D102, D202, D302 (Factory SMT) | BAT46W / D_SOD-123 | 0.3927 |
| 5 | BSS138BK,215 / C282529 | Q102, Q202, Q302, Q2, Q3 (Factory SMT) | BSS138BK / SOT-23 | 0.3345 |
| 3 | CC0805JRNPO9BN102 / C62774 | C101, C201, C301 (Factory SMT) | 1nF / C_0805_2012Metric | 0.0411 |
| 2 | CC0805JRNPO9BN200 / C105173 | C25, C26 (Factory SMT) | 20pF / C_0805_2012Metric | 0.0228 |
| 1 | CC0805KRX7R9BB333 / C107150 | C69 (Factory SMT) | 33nF / C_0805_2012Metric | 0.012 |
| 2 | CL21B103KBANNNC / C1710 | C72, C73 (Factory SMT) | 10nF / C_0805_2012Metric | 0.026 |
| 14 | CL21B104KCFNNNE / C28233 | C3, C6, C11, C20, C21, C22, C27, C40, C41, C50, C51, C66, C74, C75 (Factory SMT) | 100nF / C_0805_2012Metric | 0.5698 |

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|---|---|---|---|---|
| 5 | CL21B105KBFNNNE / C28323 | C7, C23, C61, C65, C71 (Factory SMT) | 1uF / C_0805_2012Metric | 0.2165 |
| 1 | CL21B473KBCNNNC / C53134 | C70 (Factory SMT) | 47nF / C_0805_2012Metric | 0.0173 |
| 2 | CL21B475KAFNNNE / C98195 | C8, C24 (Factory SMT) | 4.7uF / C_0805_2012Metric | 0.1216 |
| 8 | CL31A226KAHNNNE / C12891 | C4, C5, C13, C12, C60, C62, C63, C64 (Factory SMT) | 22uF / C_1206_3216Metric | 1.3552 |
| 1 | CL32B106KBJNNWE / C3844168 | C2 (Factory SMT) | 10uF / C_1210_3225Metric | 0.7145 |
| 1 | EEEFT1H470AP / C92059 | C1 (Factory SMT) | 47uF / CP_Panasonic_FT_6.3x5.8 | 0.3573 |
| 1 | ESD2CAN24DBZRQ1 / C5736151 | D20 (Factory SMT) | ESD2CAN24-Q1 / SOT-23 | 0.3486 |
| 1 | LM339LVDR / C22427665 | U7 (Factory SMT) | LM339LV / SOIC-14_3.9x8.7mm_P1.27mm | 0.5859 |
| 1 | LMR51430XDDCR / C5185863 | U1 (Factory SMT) | LMR51430XDDCR / TSOT-23-6 | 0.8479 |
| 3 | M22-2010305 / 952-2274-ND | JP101, JP201, JP301 (Hand solder) | 5V / 12V / PinHeader_1x03_P2.00mm_Vertical | 0.66 |
| 1 | MMSZ12T1G / C72599 | D3 (Factory SMT) | 12V ZENER / D_SOD-123 | 0.0723 |
| 1 | PM254V-12-10P-H85 / C492399 | J8 (Hand solder) | 2x5 SERVICE SOCKET 2.54mm / Service_2x05_P2.54mm | 0.1158 |
| 1 | RC0805FR-0713KL / C137592 | R63 (Factory SMT) | 13k / R_0805_2012Metric | 0.0055 |
| 3 | RC0805FR-0720KL / C114566 | R103, R203, R303 (Factory SMT) | 20k / R_0805_2012Metric | 0.0237 |
| 1 | RC0805FR-072K61L / C273965 | R62 (Factory SMT) | 2.61k / R_0805_2012Metric | 0.0075 |
| 3 | RC0805FR-072M2L / C137552 | R104, R204, R304 (Factory SMT) | 2.2M / R_0805_2012Metric | 0.0219 |
| 1 | RC0805JR-070RL / C96345 | R20 (Factory SMT) | 0 / R_0805_2012Metric | 0.0066 |
| 3 | RC1206FR-0710KL / C132649 | R109, R209, R309 (Factory SMT) | 10k / R_1206_3216Metric | 0.0333 |
| 6 | RT0805BRD07100KL / C122537 | R60, R65, R66, R68, R72, R73 (Factory SMT) | 100k / R_0805_2012Metric | 0.2622 |
| 1 | RT0805BRD07102KL / C865126 | R1 (Factory SMT) | 102k / R_0805_2012Metric | 0.0471 |
| 1 | RT0805BRD0712K1L / C865169 | R69 (Factory SMT) | 12.1k / R_0805_2012Metric | 0.0533 |
| 1 | RT0805BRD0713K7L / C865183 | R2 (Factory SMT) | 13.7k / R_0805_2012Metric | 0.0468 |
| 2 | RT0805BRD0720KL / C469660 | R67, R74 (Factory SMT) | 20k / R_0805_2012Metric | 0.1486 |
| 1 | SK6805-EC15 / C2890035 | D10 (Factory SMT) | SK6805-EC15 / LED_SK6805_EC15 | 0.0995 |
| 1 | SMAJ13A-13-F / C110519 | D6 (Factory SMT) | SMAJ13A / D_SMA | 0.1241 |
| 3 | SMAJ30A / C148230 | D101, D201, D301 (Factory SMT) | SMAJ30A / D_SMA | 0.4362 |
| 1 | SMBJ22A-E3/52 / C3698580 | D2 (Factory SMT) | SMBJ22A / D_SMB | 0.4671 |
| 1 | SN74AHCT1G125DBVR / C7484 | U10 (Factory SMT) | SN74AHCT1G125 / SOT-23-5 | 0.1001 |
| 1 | SRP5030TA-3R3M / C2042118 | L2 (Factory SMT) | 3.3uH / L_Bourns_SRP5030T | 2.3956 |

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|---|---|---|---|---|
| 1 | SRP7028A-8R2M / C2041815 | L1 (Factory SMT) | 8.2uH / L_Bourns_SRP7028A_7.3x6.6mm | 1.8407 |
| 1 | STM32F042K6T6 / 497-14647-ND | U5 (Hand solder) | STM32F042K6T6 / LQFP-32_7x7mm_P0.8mm | 4.34 |
| 1 | TCAN3413DR / C22433320 | U6 (Factory SMT) | TCAN3413DR / SOIC-8_3.9x4.9mm_P1.27mm | 1.0156 |
| 1 | TPS2553DBVR-1 / C111738 | U4 (Factory SMT) | TPS2553DBVR-1 / SOT-23-6 | 0.2912 |
| 1 | TPS259620DDAR / C2155773 | U9 (Factory SMT) | TPS259620DDAR / TPS2596_DDA8 | 0.8154 |
| 1 | TPS61085DGKR / C113659 | U8 (Factory SMT) | TPS61085DGKR / VSSOP-8_3x3mm_P0.65mm | 0.6006 |